
🖨️ main.cpp

```cpp

#include <iostream>

#include <iomanip>

using namespace std;

// Function prototypes

double calculateFee(int totalMinutes, int vehicleType);

string vehicleName(int type);

int main()

{

const int MAX = 5; // Maximum vehicles

string plate[MAX];

int type[MAX];

int duration[MAX];

double fee[MAX];

int count = 0;

int choice;

do {

cout << "\n===== \n";

cout << "CAR PARKING MANAGEMENT SYSTEM \n";

cout << "===== \n";

cout << "1. Calculate Parking Fee \n";

cout << "2. View All Records \n";

cout << "3. End \n";

cout << "Enter choice: ";

cin >> choice;

// ----- OPTION 1 -----

if (choice == 1) {

if (count >= MAX) {

cout << "Maximum vehicle limit reached! \n";

continue;

}

```

int eh, em, xh, xm;

cout << "\nEnter plate number: ";
cin >> plate[count];

cout << "Vehicle type (1-Car, 2-Motorcycle): ";
cin >> type[count];

if (type[count] != 1 && type[count] != 2) {
 cout << "Invalid vehicle type!\n";
 continue;
}

cout << "Enter entry time (hour minute): ";
cin >> eh >> em;

cout << "Enter exit time (hour minute): ";
cin >> xh >> xm;

// Time validation
if (eh < 0 || eh > 23 || em < 0 || em > 59 ||
 xh < 0 || xh > 23 || xm < 0 || xm > 59) {
 cout << "Invalid time format!\n";
 continue;
}

int entry = eh * 60 + em;
int exitTime = xh * 60 + xm;

if (exitTime < entry) {
 cout << "Error: Exit time cannot be before entry time!\n";
 continue;
}

duration[count] = exitTime - entry;
fee[count] = calculateFee(duration[count], type[count]);

// ----- DISPLAY RESULT -----
cout << "\n-----\n";
cout << "Plate Number : " << plate[count] << endl;
cout << "Vehicle Type : " << vehicleName(type[count]) << endl;

```

```

cout << "Entry Time : " << setw(2) << setfill('0') << eh << ":"
 << setw(2) << setfill('0') << em << endl;
cout << "Exit Time : " << setw(2) << setfill('0') << xh << ":"
 << setw(2) << setfill('0') << xm << endl;

cout << "\nTotal Parking Time: "
 << duration[count] / 60 << " hours "
 << duration[count] % 60 << " minutes\n";

if (type[count] == 1) {
 cout << "Parking Rate : RM 2.00 (1st hour)\n";
 cout << "Additional Charge : RM 1.50 per hour\n";
}
else {
 cout << "Parking Rate : RM 1.00 (1st hour)\n";
 cout << "Additional Charge : RM 1.00 per hour\n";
}

cout << "-----\n";
cout << "TOTAL PARKING FEE : RM " << fixed << setprecision(2)
 << fee[count] << endl;
cout << "=====\n";
cout << "Thank you for using our system!\n";

count++;
}

// ----- OPTION 2 -----
else if (choice == 2) {
 if (count == 0) {
 cout << "No parking records available.\n";
 }
 else {
 cout << "\n===== PARKING RECORDS =====\n";
 cout << setfill(' ');

 cout << left
 << setw(15) << "Plate"
 << setw(15) << "Vehicle"
 << setw(18) << "Duration"
 << "Fee\n";
 }
}

```

```
cout << "-----\n";
```

```
for (int i = 0; i < count; i++) {
 cout << left
 << setw(15) << plate[i]
 << setw(15) << vehicleName(type[i])
 << setw(8) << duration[i] << " mins "
 << "RM " << fixed << setprecision(2)
 << fee[i] << endl;
}
}
}
```

```
else if (choice != 3) {
 cout << "Invalid choice! Please try again.\n";
}
```

```
} while (choice != 3);
```

```
cout << "\nExiting system... Goodbye!\n";
return 0;
}
```

```
// ----- FUNCTIONS -----
double calculateFee(int totalMinutes, int vehicleType)
{
 double fee;
 if (vehicleType == 1) { // Car
 fee = 2.00;
 if (totalMinutes > 60)
 fee += ((totalMinutes - 1) / 60) * 1.50;
 }
 else { // Motorcycle
 fee = 1.00;
 if (totalMinutes > 60)
 fee += ((totalMinutes - 1) / 60) * 1.00;
 }
 return fee;
}
```

```
string vehicleName(int type)
{
 return (type == 1) ? "Car" : "Motorcycle";
}
```